CHARLEMAGNE MARC

Columbia, MD • (410) 300-4581 • chamarc1@outlook.com • chamarc1.github.io • US Citizen

TECHNICAL CORE & SKILLS

Core Languages: Python, C++, SQL, Java, Bash / Shell Scripting

Data Science & Signal Processing: NumPy, SciPy, Pandas, NetCDF4, Signal Processing, Curve Fitting

Pipeline & Automation: ETL Pipelines, SLURM (HPC), Multiprocessing, GNU Parallel, Argparse

Systems & Infrastructure: Git, Gitea, Docker, LEO OS, VS Code, YAML, Matplotlib, PIL

ENGINEERING EXPERIENCE & KEY PROJECTS

Software Engineer Intern — Earth and Space Institute (ESI) at UMBC

06/2024 – Present

- **NASA ER-2 Mission Integration:** Engineered sensor calibration and instrument software using Python and C++ within Low Earth Orbit (LEO) environments for shortwave infrared (SWIR) imagery deployed on NASA's high-altitude ER-2 aircraft during the NASA PACE-PAX field campaign.
- **Automated Data Processing:** Managed the development of automated data reduction routines, transforming raw hardware optical inputs into verified scientific data products.
- **Signal Processing & Telemetry Validation:** Built robust signal processing software utilizing quadratic curve fitting and sigma filtering architectures to automate instrument outlier removal and validate incoming telemetry data.

Lead Developer — VNIR Level-1A Quicklook Generation (ESI)

06/2026 – Present

- **Python Visualization Pipeline:** Developed an automated quicklook pipeline turning AirHARP2 VNIR Level-1A NetCDF4 data into multi-panel PNG imagery for flight coverage and quality review, extracting detector strips across four wavelength rows aligned with acquisition timestamps.
- **Flight Leg Stitching & CLI Tooling:** Implemented batch processing by time window or flight leg to stitch related granules into unified leg visualizations, supported by argument-file configurations and automated CLI test suites.

Lead Developer — SWIR Level-1A Generation Pipeline (ESI)

08/2025 – 05/2026

- **End-to-End ETL Architecture:** Architected an automated Python ETL pipeline converting raw satellite imagery and metadata into standardized Level-1A NetCDF4 format, enabling public scientific sharing across Earth science research teams.
- **Data Ingestion & Robust Parsing:** Built data ingestion workflows with argparse and custom capture-ID matching logic across heterogeneous sources, integrating strict exception handling and detailed audit tracking.
- **HPC Workflow Automation:** Implemented production batch workflow automation utilizing multiprocessing and subprocess tools within SLURM and GNU Parallel High-Performance Computing (HPC) environments for high-volume datasets.
- **Quality Assurance & Anomaly Logging:** Developed automated data-quality controls for timestamp continuity and granule completeness using YAML configurations to systematically generate real-time anomaly detection logs.
- **CLI Tooling & Geospatial Visualizations:** Wrote reusable CLI tooling, shell transformation scripts, and data visualizations leveraging Matplotlib and Pandas to clean and prepare geospatial datasets for analytical consumption.
- **CI/CD & Code Review Best Practices:** Managed version control via self-hosted Gitea and Git repositories; conducted bi-weekly structured code reviews to iteratively refine codebase reliability and maintain strict exception handling.

Developer — Flatfield Sensor Calibration Suite (ESI)

10/2024 – 08/2025

- **Matrix Computation Pipeline:** Architected automated laboratory data reduction pipelines in Python using NumPy and SciPy to perform high-performance matrix computations, applying Dark Frame and Flatfield corrections for maximum optical accuracy.
- **Telemetry Parsing & Data Reduction:** Designed automated telemetry data reduction routines to parse raw laboratory sensor telemetry and export systematic calibration matrices for downstream instrument processing.

OPERATIONAL & LEADERSHIP EXPERIENCE

Head Travel Soccer Coach — Soccer Association of Columbia (SAC)

08/2013 – Present

- **Team Leadership & Operational Management:** Direct travel soccer teams, designing tactical training programs and managing season operations; recipient of the **SAC Coach of the Year** award. Hold a **USSF National D License** and led teams to consecutive championship titles across premier regional tournaments including Baltimore Mania and Columbus Day Cup.

Campus Security Officer — UMBC Police Department

08/2017 – Present

- **Operational Logging & Leadership:** Maintained an outstanding **9+ year record** of operational logging, situational leadership, and emergency protocol coordination in a high-public-trust role.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science — *University of Maryland, Global Campus (UMGC)* — Graduated

Undergraduate Certificate in Vulnerability Assessment — *UMGC* — Completed